

TEST_LOG - verified execution record

All 15 checks passed on synthetic demo data: data/AI_A1_GXX.csv (40 rows).

Page 1 of 2. Replace this evidence after rerunning with the real lecturer-issued CSV.

1. Clean installation	./venv/bin/python -m pip check	No broken requirements found.	PASS
2. run_all.py on lecturer-style dataset	Run run_all.py on demo CSV	Pipeline completed; 40 rows loaded and outputs generated.	PASS
3. No fixed row assumptions	Run on schema-equivalent 35-row copy	Pipeline completed; data_report row_count was 35.	PASS
4. Regression loss and metrics recomputed	Independently recompute saved-model test predictions	MAE, RMSE, and R2 matched JSON; loss artifact exists.	PASS
5. Classification metrics match predictions	Independently recalculate labels and metrics	Accuracy, precision, recall, F1, and confusion matrix matched.	PASS
6. Silhouette scores k=2 through k=5	Inspect clustering_metrics.json	Scores recorded for k=2, 3, 4, and 5; selected k=5.	PASS
7. Every row receives a cluster label	Compare clusters.csv with input record IDs	All 40 input records received a non-null label.	PASS
8. Valid predict.py JSON	Run valid six-field JSON record	Valid JSON returned with all required prediction fields.	PASS

Note: These results are genuine execution evidence for the synthetic demo dataset only.

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9. Incomplete predict.py JSON	Run record containing only plot_area_ha	Readable missing-fields validation error returned.	PASS
10. No identifier/target leakage	Inspect model feature lists and clustering inputs	Only six input features used; identifier and targets excluded.	PASS
11. Scalers fitted only on training data	Inspect supervised preprocessing code	Scalers fit on x_train and only transform x_test.	PASS
12. All required files created	List isolated artifacts and models outputs	All required artifact and model files were created.	PASS
13. Delete artifacts and rerun	Remove isolated outputs and rerun pipeline	Outputs were recreated automatically.	PASS
14. Clean terminal/environment	Run with minimal clean environment	Pipeline completed successfully.	PASS
15. CSV copy with different rows	Run on 35-row schema-equivalent copy	Pipeline completed without hard-coded row assumptions.	PASS

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